Wing Theory

1 From aerofoil (2D) to wing (3D)

Aerofoil data (C_l, C_d, C_m) describe a wing of *infinite* span. Real wings are finite and their coefficients are written C_L, C_D, C_M.

- The flow is three-dimensional, with a spanwise velocity component.
- High-pressure air below the wing spills round the tip to the low-pressure upper surface. This forms the **tip vortex**, whose downwash spoils lift and creates drag.
- **Example:** the Beechcraft Baron 58 uses NACA 23015 at the root (10% thick at the tip). At $\alpha = 4°$, $Re = 6 \times 10^5$, the aerofoil gives $C_l = 0.54$, $C_d = 0.0068$. The wing's C_L and C_D are *not* these values; they must be corrected for 3D effects.

2 Wing geometry and the reference wing

Constraints: performance (cruise, loiter, take-off, landing), flying qualities, structure (spar placement), internal volume for fuel and payload, stealth, and airport span limits.
Reference wing: a trapezoidal wing with its root chord on the fuselage centreline and the same area as the planform. Its area includes the part inside the fuselage, so it is fictitious; the root aerofoil is at the centreline, not at the wing–body junction.
Parameters: reference area S, chord c, span b, aspect ratio A, thickness ratio t/c, sweep Λ (at the LE, $c/4$ or maximum thickness line), taper ratio λ, twist, dihedral.

$$A = \frac{b}{\bar{c}} \text{ (rectangular)}, \qquad A = \frac{b^2}{S} \text{ (general)}, \qquad \lambda = \frac{c_{tip}}{c_{root}}$$

$$\text{MAC} = \bar{c} = \frac{2}{S}\int_0^{b/2} c^2\, dy$$

The mean aerodynamic chord is the chord of the rectangular wing with the same forces and moments as the actual wing.

3 Aspect ratio

At equal area, a high-A wing loses less of its span to the tip vortex.

- **Lift:** higher A gives a higher lift-curve slope C_{L_α} and higher C_L.
- **Stall:** higher A stalls at a *lower* angle. Low-A wings stall later, so tails use low A to stall after the wing and keep control.
- **Induced drag:** higher A gives less induced drag.
- **Span:** for a given area, A sets the span, and so hangar space.
- **Weight:** wing weight rises with about $\sqrt{A}$, because the larger span raises the root bending moment:

$$W_{wing} \propto S_w^{0.649}\, A^{0.5}\, (t/c)_{root}^{-0.4}\, (1+\lambda)^{0.1}\, (\cos\Lambda)^{-1}$$

Trade-off: at fixed area, a very high A gives a small chord and a low $Re = \rho V c/\mu$, which can reduce C_L. A is set by a trade study of aerodynamics against weight.

Propeller aircraft	A
Homebuilt	6.0
GA single engine	7.6
GA twin engine	7.8
Agricultural	7.5
Twin turboprop	9.2
Flying boat	8.0

Jet aircraft	a	C
Trainer	4.737	−0.979
Fighter (dogfighter)	5.416	−0.622
Fighter (other)	4.110	−0.622
Cargo/bomber	5.570	−1.075
Transport	7.50	0

$A = a\, M_{max}^{\,C}$

For jets, A falls as M_{max} rises: drag due to lift matters less at high speed, so lower A saves weight.

4 Aspect ratio correction of the lift-curve slope

Downwash tilts the local flow, so the wing sees a smaller **effective** angle of attack:

$$\alpha_{eff} = \alpha - \alpha_i, \qquad \alpha_i = \frac{C_L}{\pi A e}$$

where e is Oswald's efficiency factor. Two wings with the same aerofoil give the same C_L at the same α_{eff}. Taking one of them as infinite ($A = \infty$, slope a_0):

$$C_l = a_0(\alpha - \alpha_{L=0}), \qquad C_L = a(\alpha - \alpha_{L=0}), \qquad a = \frac{a_0}{1 + \dfrac{a_0}{\pi A e}}$$

Slopes in per rad. **Example:** $a_0 = 2\pi$, $A = 8$, $e = 0.9$ gives $a = 4.92$ per rad, a 22% loss against the aerofoil.

5 Wing sweep

Sweeping by Λ keeps the thickness but lengthens the chord the flow sees, so the effective section is thinner (at 45°, t/c of 0.15 becomes 0.106). Sweep affects:

- **Lift-curve slope** (falls with sweep):

$$C_{L_\alpha} = \frac{2\pi A}{2 + \sqrt{4 + \dfrac{A^2\beta^2}{\eta^2}\left(1 + \dfrac{\tan^2\Lambda_{max,t}}{\beta^2}\right)}}$$

with $\beta^2 = 1 - M^2$, $\eta = C_{l_\alpha}/(2\pi/\beta)$. For $A = 8$ at $M = 0.8$, sweeping from 0° to 35° cuts C_{L_α} by about 25%.
- **Induced drag:** $C_{D_i} \propto 1/\cos(\Lambda - 5°)$ (experimental).
- **Critical Mach:** the section sees only the normal velocity component, so $M_{cr,wing} \approx M_{cr,aerofoil}/\cos\Lambda$; e.g. $0.7/\cos 30° = 0.81$.
- **Drag divergence Mach** M_{DD}: where C_D has risen by 0.002 above its subsonic value. Sweep raises it.
- **Weight:** $W_{wing} \propto 1/\cos\Lambda$, so it rises with sweep.

Regime	Sweep	Purpose
Subsonic	small	Place the aerodynamic centre relative to the CG; control arm on flying wings
Transonic	30°–35°	Delay the drag rise
Supersonic	45°–70°	Keep the wing inside the Mach cone
Variable	–	Best of both (F-14), but heavy and complex

6 Taper ratio

$\lambda = 1$ rectangular, $0 < \lambda < 1$ trapezoidal, $\lambda = 0$ delta. Taper affects induced drag, structural weight and ease of fabrication; it may be in planform, in thickness, or both.

- **Elliptic** loading gives the lowest induced drag ($e = 1$) but is hard to build.
- **Rectangular** is easy to build but has about 7% higher C_{D_i} and is heavier.
- Taper moves lift inboard and allows a thicker root, both of which save wing weight.
- To avoid tip stall, typical values (Raymer) are $\lambda \approx 0.4$–0.5 for unswept wings and 0.2–0.3 for swept wings.

7 Twist

Twist prevents tip stall and moves the lift distribution towards an ellipse; typically 0° to 5°.

- **Geometric twist:** the change in incidence of the section, measured from the root. Tip nose-down = **wash-out**; tip nose-up = **wash-in** (very unusual).
- **Aerodynamic twist:** the angle between a section's zero-lift line and the root's. Equal to geometric twist if the same aerofoil runs root to tip; possible with no geometric twist by changing aerofoil (e.g. symmetric root, cambered tip).
- Twist is optimal at only *one* lift coefficient, so avoid much more than 5°. About 3° usually gives adequate stall behaviour.

8 Wing incidence

The pitch angle of the wing relative to the fuselage axis (at the exposed root if the wing is twisted). It is chosen so that, at the design condition, the fuselage sits at its minimum-drag attitude (about 0° for a straight circular fuselage), and passenger cabins stay level.

Aircraft	Initial incidence
General aviation, homebuilt	about 2°
Transport	about 1°
Military	about 0°

It is finally set from wind-tunnel data.

9 Dihedral

The angle of the wings to the horizontal, seen from the front. Positive dihedral gives roll stability: sideslip during a bank creates a restoring rolling moment.

Dihedral (deg)	Low wing	Mid wing	High wing
Unswept (civil)	5 to 7	2 to 4	0 to 2
Subsonic swept	3 to 7	−2 to 2	−5 to −2
Supersonic swept	0 to 5	−5 to 0	−5 to 0

Too much dihedral causes **Dutch roll** (coupled yaw–roll oscillation). A larger fin cures it but adds weight and drag, so tail size and dihedral are chosen together.

10 Wing vertical location

	Advantages	Disadvantages
Low	Gear stowed in the wing (no pods, lower drag); carry-through structure under the cabin	Low ground clearance; destabilising in yaw, so larger fin
Mid	Lowest drag; good ground clearance; clear view (some military aircraft)	Spar cannot pass through the fuselage, so heavier
High	Fuselage low for cargo loading; engine and prop clearance with short gear; strut bracing; STOL: large flaps, less debris ingestion, no floating in ground effect	Gear in fuselage pods (weight and drag); pilot's view blocked in a turn

11 Wing tips

Tip shape changes wetted area slightly and the tip vortex strongly; a sharp edge makes it harder for air to spill round, reducing induced drag.

- **Rounded:** lets air flow round easily. **Cut-off:** sharp edges, already less drag.
- **Hoerner:** undercut lower surface; the most widely used low-drag tip. **Drooped/upswept:** raise effective span without raising actual span.
- **Aft-swept:** lower drag but higher torsional loads. **Forward-swept** (F-15): keeps the TE out of the vortex.
- **End-plate:** blocks the flow but its own wetted area adds drag.
- **Winglet:** recovers energy from the vortex; beats an equal-area span increase. Adds weight behind the elastic axis (flutter) and is optimised for one speed. Most benefit on **low-A** wings.

12 Tail functions and arrangements

"Tails are little wings" but normally work at a fraction of their lift potential. They provide **trim** (aft tail at about −2° to −3° incidence balances the wing moment; fin handles engine-out), **stability** (like the fins on an arrow) and **control** (critical cases: nose-wheel lift-off, flaps-down low speed, transonic manoeuvre; for the fin, engine-out, maximum roll rate, spin recovery).

Arrangement	Key points
Conventional	Works, lightest; about 70% of aircraft
T-tail	Heavier fin; end-plate effect allows a smaller fin; tail clear of wake and propwash; enables aft-fuselage engines. Risk of **deep stall**
Cruciform	Lifts the tailplane clear of exhaust, exposes lower rudder; less weight penalty than T-tail, no end-plate benefit
H-tail	Fins in clean air at high α or in propwash; heavier, but smaller tailplane; lowers tail height for hangars
V-tail	Less wetted and interference drag; ruddervators need a mixer; adverse roll–yaw coupling
Boom-mounted	Allows a pusher prop or engine near the CG; heavier booms

Spin recovery: at high α the stalled tailplane casts a wake upward at about 45°. The rudder must stay out of it: move the tailplane aft or up; a T-tail uncovers the rudder but risks pitch-up.

Tail geometry: areas scale with wing area (tail volume coefficient method, after the first W_0 estimate). Tailplane LE sweep about 5° more than the wing, so it stalls later and has a higher M_{cr}. Tail t/c similar to the wing, about 10% thinner on high-speed aircraft. T-tails use lower fin aspect ratios; some GA aircraft use untapered tails ($\lambda = 1$).